

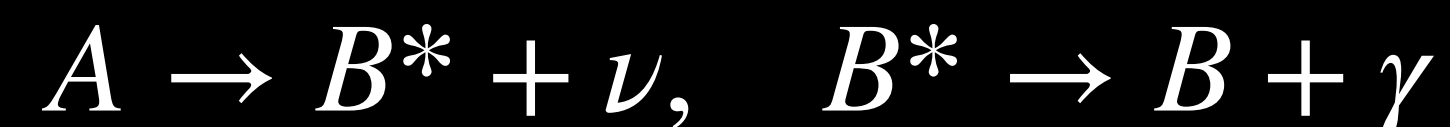






Resonant absorption as a direction filter

- ★ Main scheme:



- ★ If the excited nucleus B^* decays at rest, the emitted photon has an energy smaller than the level spacing

$$E_\gamma < E_i - E_f$$

because part of the energy goes into the recoil of nucleus B .

- ★ Therefore, such a photon cannot be resonantly absorbed by another nucleus B at rest: it is simply not energetic enough.

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